

AI TRACEFINDER

Document Source Identification using Scanner Pattern

Milestone – 1 Report

Infosys Springboard Internship

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1. Introduction

With the increasing use of scanned documents for official and legal purposes, ensuring the authenticity and origin of scanned images has become an important challenge. Every scanner device introduces unique noise patterns and artifacts during the scanning process, which can be exploited to identify the source scanner.

The goal of **AI TraceFinder** is to identify the scanner device used to produce a given scanned document by analyzing these scanner-specific characteristics. This milestone focuses on **dataset preparation and preprocessing**, which form the foundation for robust scanner identification.

2. Objective of Milestone-1

The objectives of Milestone-1 are:

- To collect scanned document datasets from multiple scanner devices.
- To organize the datasets according to scanner models.
- To create accurate labels for each scanned image.
- To preprocess images by converting them into a uniform format suitable for machine learning.

3. Dataset Description

The datasets used in this project consist of scanned documents collected from different scanner devices and sources:

- **Flatfield:**
These images are scans of blank or uniform pages. They capture scanner-specific noise patterns and sensor artifacts, which are crucial for forensic analysis.
- **Official:**
This dataset contains real-world official documents scanned using multiple scanners. It represents authentic usage scenarios.
- **Wikipedia:**
These scans include documents with diverse layouts, formats, and textual structures, improving dataset diversity and robustness.

Each dataset is organized into subfolders representing individual scanner models such as **Canon, Epson, and HP**, ensuring clear identification of the source scanner.

4. Dataset Organization and Labeling (Week-1)

4.1 Folder Structure

The dataset is structured as follows:

```
Datasets/  
├ Flatfield/  
├ Official/  
├ Wikipedia/  
└ <Scanner_Model>/  
    └ image files
```

Each subfolder corresponds to a specific scanner model, and the folder name is used as the label for all images contained within it.

4.2 Label Creation

An automated Python script was developed to generate a **labels.csv** file. The CSV file maps each image to its corresponding scanner model and dataset type.

Fields in labels.csv:

- label – Scanner model name
- filepath – Absolute path to the image
- dataset – Dataset category (Flatfield / Official / Wikipedia)

This automated approach ensures consistency, reduces manual errors, and supports scalability.

5. Image Preprocessing (Week-2)

To prepare the dataset for feature extraction and model training, all images were preprocessed using the following steps:

5.1 Grayscale Conversion

All images were converted to grayscale to emphasize scanner noise patterns while reducing color-related complexity.

5.2 Image Resizing

Each image was resized to a uniform resolution of **512 × 512 pixels**. This ensures consistency across all inputs and compatibility with machine learning algorithms.

5.3 Junk File Handling

Hidden macOS metadata files (e.g., `._filename.tif`) were automatically detected and skipped to prevent invalid image reads.

5.4 Processed Dataset Storage

Preprocessed images were stored in a separate **Processed** folder while maintaining the original dataset structure:

```
Processed/  
├─ Flatfield/  
├─ Official/  
├─ Wikipedia/  
└─ <Scanner_Model>/  
    └─ processed images
```

This separation preserves the original raw data and enables clean experimentation.

6. Verification and Validation

An automated verification script was used to validate preprocessing results across the entire dataset.

The verification confirmed:

- ☒ All images are grayscale
- ☒ All images have consistent size (512 × 512)
- ☒ No junk or hidden files exist in the processed dataset
- ☒ All processed images are stored in the correct directory

Successful verification indicates that the dataset is clean, consistent, and ready for feature extraction.

7. Outcome of Milestone-1

By the end of Milestone-1, the following outcomes were achieved:

- A well-organized and labeled scanned document dataset.
- A clean, preprocessed dataset suitable for forensic analysis.
- Automated scripts for reproducibility and scalability.

- Verified readiness for subsequent feature extraction and modeling tasks.

8. Conclusion

Milestone-1 successfully established a strong foundation for the AI TraceFinder project. Through systematic data collection, accurate labelling, comprehensive preprocessing, the data is now in an optimal form for further analysis. The next milestone focus on **feature extraction and scanner fingerprint analysis** to enable effective scanner identification using machine learning.